

Article

Adoption of Conceptual Model for Smartphones among Older People

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Abstract: A critical issue is the acceptance of current modern smartphone technology by older people. Smartphones are inventions that presently offer significant advantages to help individuals. However, around the world, every older user wants to perform tasks that are more user-friendly, and near their personal preferences. Considering these factors, this research aimed to explain, identify, and examine multiple external factors and usability for acceptance, adoption, and usage of smartphones by the 45-year-old-and-above population. To achieve the research objective, a new conceptual model, the smartphone acceptance model for older people, was proposed; it was principally based on Technology acceptance model (TAM), in order to identify the essential aging factors that may have a significant influence on smartphone usability, usage, acceptance, and adoption among older people. The data collection was conducted through a questionnaire that was administered to 240 persons above 40. Overall findings showed that multiple external factors such as comfortability (CA), social influence (SI), perceived enjoyment (PE), fear of failure (FOF), perceived ease of use (PEU), perceived usefulness (PU), behavioral intention (BI), compatibility (COM), and attitude towards use (ATU) had significant impacts in the adoption of smartphones among older persons. For hypothesis testing, the collected data were additionally analyzed, and it was found that of the ten hypotheses, nine were positively significant overall. SEM-PLS and SPSS were used to analyze the collected data, and the findings were based on the supported hypotheses. This research contribution can be seen in terms the identification and examination of external factors that hinder or empower acceptance of smartphones among the older population. Moreover, it will help developers and manufacturers of smartphones concerning aspects that are appropriate for smartphone and app design, particularly for older users.

Keywords: older persons; technology acceptance model; smartphone adoption



Citation: Yousaf, A.; Iqbal, M.W.; Arif, M.; Jaffar, A.; Brezulianu, A.; Geman, O. Adoption of Conceptual Model for Smartphones among Older People. *Appl. Sci.* **2022**, *12*, 12703. <https://doi.org/10.3390/app122412703>

Academic Editors: Paolino Di Felice and Juan Pavón

Received: 18 October 2022

Accepted: 6 December 2022

Published: 11 December 2022

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1. Introduction

Cell phone deployment is a phenomenon that has eliminated all gender and age limits, globally. Technological developments provide comfort; however, they also bring some unfriendly impacts [1]. People of every gender and age use mobile phones, but there are always differences in usage between older and younger generations [2]. Smartphones offer people ways to become involved in social communications and industry. The smartphone allows individuals to share information and knowledge, flexibly and rapidly. Currently, a significant part of the world uses smartphones because they have outpaced the Internet, personal computers (PCs), and the social networking boom. The older a person becomes, the more challenging it is to accept and adopt technology. The concern is that older people are often not adopting innovative mobile technologies, or do not know how to use them.

Smartphone technology is so successful that its usage multiplies by tens of thousands daily [3].

The level of comfort individuals that have in interacting with advanced devices, accomplishing ideal execution concerning effectiveness, with their adequacy, and users' satisfaction, is considered usability [4]. Smartphones need to provide extreme usability to lessen the generation gap. Usability enables smartphones to replace many existing technologies such as calculators, digital cameras, and personal computers. It has not been confirmed that older individuals discard innovation more than individuals of different ages; older people, just as any other persons, accept and embrace innovation when it meets their requirements and desires [5]. The elderly can be relied upon to avail alternate ways to deal with the creation of smartphone technologies than younger individuals, due to their maturity.

The world is aging rapidly; however, for the older populace, the ratio of smartphone adoption is relatively low. Older individuals seem to have been an ignored group in the process of smartphone design. The low adoption of smartphones by older users in every developing country may be a result of the learning troubles they experience, where older people are the leading growing age group that supports the acceptance, adoption, and usage of smartphones, which are generally used to perform fundamental functions. Individuals' particular characteristics influence their potential, and bring impediments to smartphone use, as well as endorsement and adoption.

Prior studies suggested that older individuals seem to have been an ignored group in smartphone design in many countries [6]. Thus, there is a dire need to design technology for older persons that meet their requirements in different domains [7]. For instance, older users' low adoption of smartphones may be because of the learning troubles they experience. It is essential to consider individual attributes, and how these qualities impact technology adoption, in order to limit the digital divide between younger and older people. However, there is a lack of empirical studies addressing the issue of smartphone adoption among the elderly. This research aimed to fill the gap in the previous research by investigating and recognizing the factors that limit smartphone technology adoption by older people. The motivation of this research was to recognize the reasons that impact the acceptance, use, and adoption of smartphones by older people.

This research determined to discover and recognize elements that affect smartphone adoption among older people; furthermore, it will add to an understanding of older people's motivations for adopting smartphones. Specifically, the research aimed to examine construct measures: social influences, perceived usefulness, compatibility, perceived ease of use, fear of failure (atychiphobia), perceived enjoyment, technology awareness, attitude towards service, and comfortability on smartphone acceptance among older people, using the technology acceptance model [8]. This research is necessary to industry, academia, and smartphone users. In particular, it upgrades perceptions in terms the use and adoption of smartphones among the older adult population. The findings of this study will help smartphone manufacturers and application engineers understand that it is essential to recognize the affecting causes that influence smartphone adoption, with a specific end goal to develop their products (smartphones) and services accordingly. Smartphones can enhance the personal satisfaction and quality of life of older people.

2. Literature Review

Mobile phone usage has increased rapidly in recent years. A few regions of the world have realized the rapid utilization and great diffusion of mobile telephony. The use of mobile phones is a phenomenon that has eliminated age and gender boundaries, globally [8]. Currently, the majority of cell phones are known as "smartphones", because they offer more compelling computing power and connectivity than normal cell phones. Recently, smartphones stand out as being among the most prominent and advanced purchase items on the planet. Technology innovation has become more versatile, and makes individuals more associated. There continue to be discussions and concerns about the distinction

between native and non-native clients of digital media [9]. The number of people having a smartphone is gradually increasing, regardless of the reality that most do not utilize a large portion of these advanced functions.

Older people usually fall behind youngsters when it comes to technology adoption. Older people's fears are generally associated with death expectancy or health problems of their spouses or families. Furthermore, older adults are concerned about their everyday life in terms of financial survival [10,11]. When using the interface of a mobile phone, age is an essential factor that influences user performance. Several adoption theories incorporated age, and frequently related it to the use and intention towards smartphone technology. It was found that the core indicator is the age for proper deployment of mobile facilities [12]. Age performs a noteworthy part in the acknowledgment of innovations [13].

Prior research has significant findings that indicate that few grown-ups show acceptance and interest in technology adoption [14]. Although it is critical to think about the ideas and attitudes of older people regarding innovation, more research is necessary to consider why older people hold such a state of mind, views, and opinions of technology. Specifically, evaluating individual characteristics of older people (such as personality, competence, etc.) may be prescient of technology acceptance and adoption [15,16]. The challenges in the adoption of smartphones faced by elderly users have been investigated in many empirical studies. For example, Juan Carlos suggested that older individuals are quite under-considered in the innovation domain, and are perceived as "non-technological," yet they acknowledge innovation as long as it addresses their necessities; however, the elderly generally maintain some distance from innovation technologies, and never absolutely accept them in the same way as youngsters.

The needs and requirements of older users have generally not been appropriately considered [17]. In the case of smartphones, older people's requirements consists of 10 factors: identification of a particular function, readability, individual concern, multi-tap, soft key issues, awareness and attractiveness, hardware capability, touch screen interface, connectivity, social impact, and learning [18]. Roupia recommended that older users have an inherent fear of new technologies, and they use technology less than younger adults (Czaja et al., 2006) [19].

Furthermore, the most important part of this research is to propose a conceptual framework. According to Maxwell, a conceptual framework describes, either in descriptive form or graphically, the important things that must be considered to be crucial reasons, variables, or ideas, and the assumed relationships between them [20]. Many external factors have been purposed in the adoption of technology, through different models in the literature; for example, Karoly Bozan (UTATU) [21] discussed the social influence in the adoption of health IT patterns; in other words, how much an individual notices that other people essential to him/her (for example, friends, family, or other close ones) trust that she/he should adopt a new technology such as a smartphone. Similarly, he described behavioral intention as the level to which an individual has consciously planned to further utilize a device or technology in the future.

According to Davis's study, PEOU alludes to an individual's expectancy of the effort essential to use a system or an application system. Perceived usefulness means an individual's opinion that a system or application will enhance job performance, attitude towards use; it refers to ways of thinking about the usage of anything [22]. The Technology acceptance model 3 (TAM 3) refers to an individual's amount of enjoyment of utilizing a particular technology, besides any performance outcomes caused by the technology or system used. Fear of failure leads a person to fear that they are not good enough to attempt certain activities. From the literature, it is clear that explanations of these factors are imperative and crucial for research, as these permit hypothesis development, and provide a foundation for the research questions in the next section [23,24].

3. Research Methodology

3.1. Hypothesis Development

Several adoption and acceptance theories incorporate age, and frequently relate it with smartphone technology use and intention. These theories include the Theory of planned behavior (TPB), Technology acceptance model (TAM), theory of planned behavior, and unified theory of acceptance and technology, which discuss the acceptance of technology in multiple contexts [25]. Hur et al. and Gafni and Geri also play a noteworthy part in acknowledging innovation [26]. In Davis, 1986, TAM was proposed to determine why a person rejects or accepts information technology. It stands out amongst the most important and cited models that clarify the elements that control IT acceptance.

According to Davis, perceived usefulness is an individual's possibility that utilizing a particular system or application will boost someone's life or job performance. Meanwhile, PEOU can explain how much an individual considers that using a specific device, technology, or, system would be effortless. Attitude towards use (ATU) is a destructive or constructive assessment of individuals, items, activities, events, concepts, or anything in the surroundings. In general, attitudes are about having negative or positive interpretations of a person, event, thing, or place [27].

3.2. Relationship between External Variable and ATU

The study clearly shows that the ATU (attitude toward use) is directly involved in the user's attraction to using a certain information system. Comfortability is a feeling of physical or mental ease, frequently categorized as an absence of hardship or how much something or somebody is comfortable. Thus, we can create a positive association between comfortability and smartphone adoption. For this hypothesis, we can predict that smartphone acceptance and adoption is influenced by gender and age differences.

Hypothesis 1. *Comfortability significantly influences the attitude towards smartphone acceptance, adoption, and service.*

Pai and Huang and Lopez-Nicolas et al. stated that social influence is an immediate factor in the intention to technology usage. Furthermore, they described that social impact is the level of how an individual trusts that significant others require them to utilize technology. A hypothesis can be created that the attitude to using a smartphone can be affected by social influence.

Hypothesis 2. *Social influence significantly influences the attitude towards smartphone acceptance, adoption, and service.*

The relationship of enjoyment with ATU is essential for older adults' intention toward smartphone adoption [28]. It represents an individual's amount of pleasure in utilizing a particular technology, besides any performance outcomes caused by technology or system use.

Hypothesis 3. *Perceived enjoyment has a significant influence on the attitude towards smartphone acceptance, adoption, and use.*

Fear of failure has a positive influence on attitudes towards actual smartphone usage. A persistent fear of failure makes people believe that they are not good enough to attempt certain activities. Therefore, older people can be influenced by fear of failure, which means that they would feel fear of failure or defeat and do not accept their weaknesses that directly affect smartphone acceptance and adoption. Thus, a hypothesis can be generated as follows:

Hypothesis 4. *Fear of failure (Atychiphobia) has a significant influence on the attitude towards smartphone acceptance, adoption, and use.*

Kurniawan stated that older people have different finger dexterity, working memory, hearing, and visual sensitivity compared to youngsters; they may have difficulties using phones with small text, screens, buttons, and with complex functionalities. Thus, the ability to use technology alludes to being aware of an innovation that is currently becoming popular, and is being promptly acknowledged in the market and industry [29]. We can create a hypothesis that technology awareness positively impacts smartphone users in older persons.

Hypothesis 5. *Technology awareness has a significant influence on the attitude towards smartphone acceptance, adoption, and use.*

3.3. Relationship between PEOU and AT

PEOU has been stated to affect advanced technology acceptance and adoption intensely. Other than its direct influence, ease of use is a significant parameter of usefulness for older people [30]. According to the technology acceptance model, individual attitudes toward using a system are directly influenced by perceived ease of use, according to these studies [31]. Melenhorst and Alsamydai suggested that perceived ease of use is crucial in a user's attitude toward a system [32]. The more consumers think the system is designed to be useful, the more positive their attitude toward technology becomes. Thus, attitudes will be more positive if older people feel that the smartphone is easy to use.

Hypothesis 6. *Perceived ease of use has a significant influence on the attitude towards smartphone acceptance, adoption, and use.*

3.4. Relationship between PEOU and PU

Perceived usefulness and ease of use are of great significance for older adults in using technology; as per Davis's study, perceived ease of use positively impacts perceived usefulness. J. Alsamydai et al. and Su et al. stated that perceived ease of use has a favorable effect on perceived usefulness, since it allows a person to feel at ease performing a task with the least amount of effort; thus, we can create a hypothesis in the following statement:

Hypothesis 7. *Perceived usefulness has a significant influence on the attitude towards smartphone acceptance, adoption, and use.*

Gelderblom et al. stated that behavioral intention is the level to which an individual has consciously planned to utilize a device or technology in the future. K. Chen et al. further discussed that behavioral intention is the degree of the probability of an individual employing the IS. A person's attitude expects their purpose and intention to outline the actual behavior (AU). Furthermore, these determinants are used to describe older people's acceptance of mobile devices. A hypothesis for this variable can be the following:

Hypothesis 8. *Behavioral intention significantly influences the attitude towards smartphone acceptance, adoption, and use.*

Both perceived usefulness and PEOU control attitude towards use (AT). The influence of PEOU on attitude toward use is also mediated by PU. Actual usage is directly impacted by BI, which AT and PU predict. Both the attitude towards use and perceived usefulness (PU) affect BI, and are also affected by perceived ease of use (PEOU). Perceived usefulness positively influences perceived ease of use, and equally affects attitudes.

Hypothesis 9. *Attitude towards use has a positive influence on actual smartphone usage.*

Compatibility alludes to how much the adoption of technology innovation is dependent on the current needs, experience, and values of potential adopters of the innovation [33]. Chen et. al stated that compatibility is an essential attribute that should be available if someone decides to adopt, accept, and use technological innovation. From this perspective, the association of smartphone utilization in the workplace with a person's way of work and lifestyle represents compatibility [34,35].

Hypothesis 10. *Compatibility has a significant influence on the attitude towards smartphone acceptance, adoption, and use.*

From the extensive literature review, a framework of this study is shown in Figure 1. It was assumed that the external variables (comfortability (gender, age), social influence, perceived enjoyment, fear of failure, technology awareness, behavioral intention) and mediating variable (attitude towards use) would influence the dependent variable (actual smartphone use). The attitude towards using a smartphone is a significant mediator between the external independent variables and smartphone adoption [36]. Furthermore, the study proposed that the external, independent variables would predict the attitude towards use, which, in turn, would signify smartphone acceptance and adoption. This indicates that if an individual perceives smartphones to be more beneficial, then this will cause individuals to have more encouraging attitudes toward smartphones, and consequently impact greater smartphone adoption.

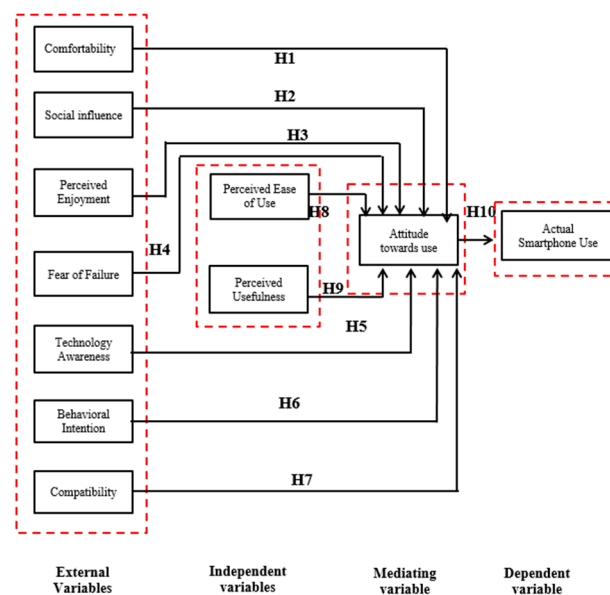


Figure 1. Conceptual framework—Smartphone Acceptance Model for Older People (SAMOP).

4. Methodology

This research investigated the effect of external factors on dependent variables, in order to examine how they affect smartphone adoption by older people. A questionnaire approach was applied to answer the research problem. The target population was the 45-year-old-and-above population. For data analysis, SmartPLS 3 (SmartPLS, Oststeinbek, Germany) and SPSS version 20 (IBM, New York, U.S.) were adopted to inspect the reliability and validity of the research questionnaire, as well as analyze the relationships between multiple independent and dependent construct measurements. The questionnaire was piloted with 240 people aged 45 and above, of which 228 responses were collected. The dependent variable of the study is actual smartphone use (adoption and acceptance). Independent variables comprised comfortability, social influence, perceived enjoyment, behavioral intention, fear of failure, perceived ease of use, technology awareness, perceived usefulness, and attitude toward use. Furthermore, the mediating role of attitude towards

use was also tested, in order to check how the attitude towards use affects the dependent variable (actual smartphone use).

4.1. Questionnaire

In this research, a questionnaire was used as a research instrument. Mai and J.E. and Galesic and Bosnjak described a collection of written questions with a choice of answers used by respondents to collect their responses and fulfill the purpose of the research [37,38]. The questionnaire was portioned into three sections. Part A included questions designed to collect respondents' demographic information that consisted of age, gender, employment status, and education stage or level. In Part B, questions were intended to gather information from the participants (smartphone users) about attitudes toward smartphone use that are influenced by external variables. In Part C, questions were composed to gather the data from respondents who were not willing to use a smartphone.

All of these questionnaire sections were answered using a Likert-type scale, in which participants were asked to choose from the following: 1 = SD (Strongly Disagree), 2 = D (Disagree), 3 = N (Not Sure), 4 = A (Agree), and 5 = SA (Strongly Agree). For the reliability and internal consistency measures, Cronbach's alpha (CA) of 0.7 was used as a standard tool in which values greater than 0.70 depicted acceptable reliability [39,40].

4.2. Participants

The population of this research involved people aged 45+ years old, both male and female. The target respondents selected were adults aged 45 and above who were using or not using a smartphone. They included working adults, non-working adults, and retirees. The population sampled in this research consisted of 240 individuals with varied ages, education, and employment backgrounds, from which 228 responses were received. The probability technique for sampling was used. The study proceeded further with data analysis using SmartPLS and SPSS to inspect the reliability and validity of the research questionnaire, and the relationship between multiple independent and dependent construct measurements.

5. Results and Discussion

The population sampled in this research consisted of 240 individuals with varied ages, education, and employment backgrounds; 228 complete replies were obtained. Their smartphone experiences were presented in a descriptive analysis, and by multiple regressions using SPSS version 20.

To check the internal reliability or consistency, the composite reliability was examined, where the value should be higher than 0.70 for satisfaction. For this purpose, SmartPLS 3 was used to accomplish numerous tests, and the outcomes are presented in Table 1. The findings in Table 1 demonstrate that the overall values of composite reliability are higher than 0.7, which indicates that the data fulfilled the internal consistency reliability test. Convergent validity was the second test applied to this proposed model, and average variance extracted (AVE) was used for this. As shown in Table 1, the data satisfied convergent validity because the smallest AVE value was 0.981, which is acceptable.

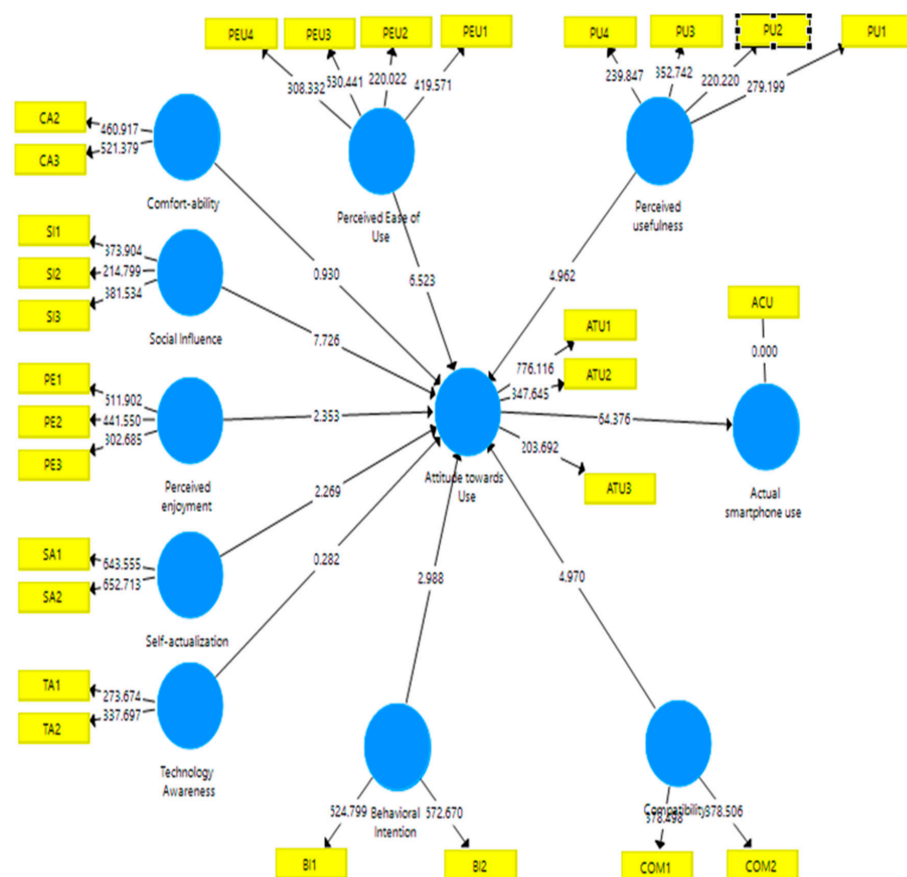
Indicator reliability was the third test used; it reflected the factor loading from each indicator. In this research, the R-squared (R²) of 0.736 was significant for the attitude towards using smartphones. On the other hand, for smartphone actual use, the R-squared was 0.991, and was considered significant. Discriminant validity was the last test in the reliability analysis. Primarily, all of the cross-loadings of an indicator have to be smaller than that indicator's loadings; as can be seen in Table 1, all indicator loadings were greater than all of their cross-loadings. Moreover, the highest squared correlation of any construct with some other latent construct must be smaller than the AVE of each latent construct. In other words, the latent construct correlations and the AVE square roots must be associated. The R²s of average variance extracted are shown diagonally in Table 1, where every square root value is greater compared to any other latent cross-correlations. Thus, the reflective measurement test satisfied this proposed model.

Table 1. Factor correlations, loadings, average variances extracted, R2s of AVE, Cronbach's alphas, and composite reliabilities of developed research model.

	Cross-Correlations										Item Loadings	AVE > 0.50	CR > 0.07	R2	CA > 0.07
	CA	SI	PE	FOF	TA	PEU	PU	BI	COM	ATU					
CA	0.9137	-	-	-	-	-	-	-	-	-	0.891–0.927	0.963	0.981	-	0.964
SI	-	0.9247	-	-	-	-	-	-	-	-	0.915–0.934	0.950	0.983	-	0.973
PE	-	-	0.954	-	-	-	-	-	-	-	0.946–0.964	0.969	0.990	-	0.982
FOF	-	-	-	0.964	-	-	-	-	-	-	0.928–1	0.964	0.982	-	0.961
TA	-	-	-	-	0.934	-	-	-	-	-	0.868–1	0.934	0.966	-	0.926
PEU	-	-	-	-	-	0.943	-	-	-	-	0.910–0.975	0.957	0.989	-	0.985
PU	-	-	-	-	-	-	0.941	-	-	-	0.912–0.972	0.956	0.989	-	0.984
BI	-	-	-	-	-	-	-	0.983	-	-	0.965–1	0.982	0.991	-	0.981
COM	-	-	-	-	-	-	-	-	0.963	-	0.922–0.973	0.963	0.981	-	0.959
ATU	-	-	-	-	-	-	-	-	-	0.923	0.886–0.975	0.949	0.982	0.736	0.973
ASU	-	-	-	-	-	-	-	-	-	-	-	-	-	0.991	-

5.1. SmartPLS Results for Structural Model

To evaluate the indicator's significance of this model, bootstrapping was employed. The number of bootstrap samples that SmartPLS typically creates is about 5000. For this research, SmartPLS selected the samples 5000 times, randomly from 200 cases, to provide the results shown below in Figure 2.

**Figure 2.** SmartPLS bootstrap results.

The t-value is known as the indicator's weight, and can be obtained from bootstrapping results. The numerical values on the lines between indicators, as shown in Figure 3, represent the t-values. The t-values are used to indicate the path's significance (p). For a

two-tailed test, the critical t-value significance level = 1% or 0.01 is equal to 2.58; 0.05 or 5% is equivalent to 1.96; and the level 10% or 0.10 is equal to 1.65.

From the hypothesis analysis shown in Table 2, it was found that out of the total nine hypotheses, nine are supported, which were comfortability (H1), social influence (H2), perceived enjoyment (H3), fear of failure (H4) perceived ease-of-use (H6), behavioral intention (H7), perceived usefulness (H8), compatibility (H9), and attitude towards use (H10); all of these resulted in intense significance levels, with coefficients = 0.930, 7.726, 2.353, 2.269, 6.523, 2.988, 4.962, 4.970, and 64.37, respectively. Technology awareness was not supported by this model for the 45–65+ age groups.

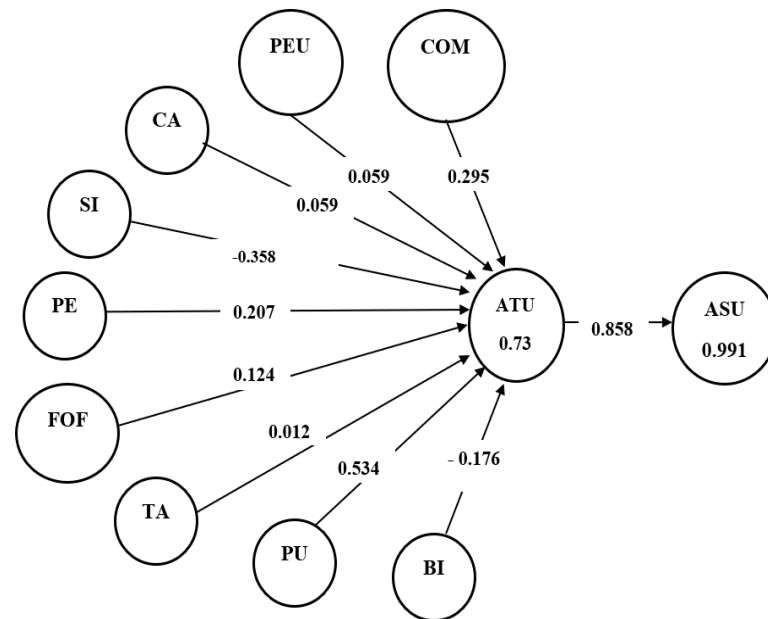


Figure 3. Hypothesis results on construct model.

Table 2. Research hypotheses, t-values, path coefficients, and hypothesis testing.

Research Hypothesis	Standardized Path Coefficients	t-Value	Hypothesis Testing (Support)
Comfortability → Attitude towards Use	0.059	0.930	Yes
Social Influence → Attitude towards Use	−0.358	7.726	Yes
Perceived Enjoyment → Attitude towards Use	0.207	2.353	Yes
Fear of Failure → Attitude towards Use	0.124	2.269	Yes
Technology awareness → Attitude towards Use	0.012	0.282	No
Perceived Ease of Use → Attitude toward Use	0.419	6.523	Yes
Behavioral intention → Attitude towards Use	−0.176	2.988	Yes
Perceived Usefulness → Attitude towards Use	0.534	4.962	Yes
Compatibility → Attitude towards Use	0.295	4.970	Yes
Attitude towards use → Actual smartphone Use	0.858	64.376	Yes

5.2. Hypothesis Testing and Comparison

The findings contained details of validity and reliability tests that consisted of explanations about the formative, reflective measurement, and structural equation model analysis. For hypothesis testing, the collected data were additionally analyzed and constructed so that of the overall ten hypotheses, nine were positively significant. Moreover, the developed model predicted 74% of attitudes towards the use of a smartphone, and 99% of smartphone actual use. Hypothesis results are presented in Figure 3.

H1: The analysis findings showed a positive association between comfortability and smartphone adoption. This hypothesis predicted that smartphone acceptance and adoption

are influenced by gender and age differences. Both gender and age affected the attitude towards actual smartphone use. The findings showed that the first hypothesis was not supported by older people. Consequently, it can be agreed that age and gender have no positive effect on attitude towards use, which shows that as people get older, their actual smartphone use is affected. With increasing age, adoption of smartphones is influenced, and they do not feel comfortable.

H2: Enjoyment was examined as an essential interpreter for older adults' intentions towards smartphone adoption. The results recommend that increasing and refining individual smartphone enjoyment could emphatically facilitate their aim to accept and use smartphones. This hypothesis was supported. Therefore, older people are influenced by entertainment, which means that when they enjoy using smartphones, their attitude changes toward acceptance and adoption of smartphones.

H3: This hypothesis showed that the attitude to use smartphones was affected by social influence. However, this hypothesis was positively supported. Thus, older people are influenced by society.

H4: This hypothesis had a positive influence on attitudes toward actual smartphone usage. Therefore, older people are influenced by a fear of failure, which means that they have a fear of failure or defeat and do not accept their weaknesses, which directly affects smartphone acceptance and adoption.

H5: It was expected for this hypothesis that the better awareness old adults have about technology, the further they aim to use technology. However, the findings for this hypothesis were not positively supported for older people. Hence, it can be suggested that older people have little awareness of smartphone innovation, but are still not interested in using smartphones.

H6: The results showed that this hypothesis was supported. If a smartphone is seen as simple to use and learn, older people are inclined to adopt it. Therefore, a higher PEOU increases the intentions of older people toward smartphone acceptance and adoption.

H7: If older people find that smartphones provide benefits, then they tend towards smartphone adoption. The results show that this hypothesis was supported. There is a strong positive impact of perceived usefulness on smartphone adoption.

H8: The results showed that actual smartphone use was affected by behavioral intention. Thus, this hypothesis was significantly supported.

H9: The results of the hypothesis anticipated that the more the smartphone is compatible with the user's work and lifestyle, the more they want to accept the smartphone innovation. This hypothesis was supported. Smartphones may not yet be compatible or are less compatible with their lifestyles, which has a stronger effect on smartphone adoption by older people.

H10: This hypothesis was supported. Therefore, it shows that attitude influences actual smartphone use.

The findings of this study showed that the smartphone acceptance model constructs, namely comfortability, social influence, perceived enjoyment, fear of failure, perceived ease of use, behavioral intention, perceived usefulness, compatibility, and attitude toward use had a significant positive influence on elderly users' actual smartphone use; these results are consistent with those of various studies. Results revealed a positive influence of perceived ease of use and perceived usefulness on elderly users' use of the smartphone. This outcome is consistent with the results from other prior studies (e.g., [40,41]). In addition, social influence, perceived enjoyment, and fear of failure had a great impact on attitudes towards the use of smartphones by elderly users, a finding which is consistent with the study.

Furthermore, compatibility and behavioral intention also had a positive influence on elderly users' attitudes toward the adoption of smartphones, and these were consistent with the findings of earlier studies. This study also found an insignificant influence of technology awareness on elderly users' attitudes towards the use of smartphones. This finding is in line with the results reported in the work of [42].

6. Conclusions

A conceptual research model was proposed based on previous research that was related present theoretical theories, and on the application of multiple qualitative and quantitative methods to attain professional views. The collected data were utilized for the developed research model to discover the significant influences that are related to older people's smartphone acceptance and adoption in developing areas. The conducted research delivers valued knowledge into the acceptance and adoption of smartphones. The findings of this paper contained detailed validity and reliability tests that consisted of explanations about the formative, reflective measurement, and analysis of the structural equation model. For the hypothesis testing, the collected data were additionally analyzed and constructed so that of the overall ten hypotheses, nine were positively significant. Moreover, the developed model predicted 74% of attitudes toward the use of a smartphone, and 99% of smartphone actual use.

Our study provides many theoretical and practical implications for the existing body of knowledge. First, although multiple studies observed the influence of technology on older persons and their acceptance behavior, very few studies examined the external factors that affect smartphone adoption by older adults. Our study extended the literature by proposing a technology acceptance model. The second important implication is related to the benefit of overcoming any issues in the previous literature on the attitudes towards using smartphones via an analysis of important concepts from the DOI, TAM, and UTAUT theories. The third prominent contribution to this research is the importance of considering older adults in smartphone design. Smartphone manufacturers should design devices that are easy to use for older people, in order to enhance smartphone adoption among them. Application developers and smartphone manufacturers also receive benefits from this research, as it indicates that the shareholders must pay more consideration to age groups. Moreover, societies or communities should be formed for older people, in order to offer learning and training about novel innovative technologies, and to assist them in developing an understanding that can stimulate them to use advanced technologies. Businesses that are related to older people should consider offering older people compatible applications such as online shopping applications, or applications related to health information for older people.

Future studies must include an understanding of how older people utilize smartphones. From a life expectancy point of view, older adults are not too old to gain useful knowledge. There is a chance to initiate learning environments for older people with technology awareness. Furthermore, to generalize the results, more research is mandatory for testing the research model with diverse members from multiple regions and cultures.

Author Contributions: Conceptualization, A.Y.; Methodology, A.Y. and M.W.I.; Formal analysis, A.Y. and A.B.; Investigation, O.G.; Resources, M.A. and O.G.; Data curation, M.W.I. and A.B.; Writing—original draft, A.Y.; Writing—review & editing, M.W.I.; Supervision, M.W.I. and M.A.; Project administration, O.G.; Funding acquisition, A.J. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Contract No. 2665/07.02.2022 of Greensoft/University of Medicine and Pharmacy “Grigore T. Popa”, Iasi, Romania, project name: Living Lab.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Acknowledgments: We are thankful to researchers and collaborators for their support.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Wang, K.H.; Chen, G.; Chen, H.-G. A Model of Technology Adoption by Older Adults. *Soc. Behav. Personal. Int. J.* **2017**, *45*, 563–572. [\[CrossRef\]](#)
2. Bracken, C.C.; Yang, H.; Pettey, G. What I Love about Technology: Older Adults and Mobile Communication Technologies. *Stud. Media Commun.* **2015**, *3*, 127–133. [\[CrossRef\]](#)
3. Holden, H.; Rada, R. Understanding the Influence of Perceived Usability and Technology Self-Efficacy on Teachers' Technology Acceptance. *J. Res. Technol. Educ.* **2011**, *43*, 343–367. [\[CrossRef\]](#)
4. Feroz, I.; Ahmad, N.; Iqbal, M.W. Usability Based Rating Scale for Mobile Health Applications. In Proceedings of the International Conference on Engineering and Emerging Technologies (ICEET), Lahore, Pakistan, 21–22 February 2019; IEEE: New York, NY, USA, 2019; pp. 1–7.
5. Holzinger, A.; Searle, G.; Nischelwitzer, A. On Some Aspects of Improving Mobile Applications for the Elderly. In *Universal Access in Human Computer Interaction. Coping with Diversity*; Stephanidis, C., Ed.; Lecture Notes in Computer Science; Springer: Berlin/Heidelberg, Germany, 2007; Volume 4554, pp. 923–932.
6. Briede-Westermeyer, J.C.; Pacheco-Blanco, B.; Luzardo-Briceño, M.; Pérez-Villalobos, C. Mobile Phone Use by the Elderly: Relationship between Usability, Social Activity, and the Environment. *Sustainability* **2020**, *12*, 2690. [\[CrossRef\]](#)
7. Abdulrazak, B.; Malik, Y.; Arab, F.; Reid, S. PhonAge: Adapted SmartPhone for Aging Population. In *Inclusive Society: Health and Wellbeing in the Community, and Care at Home*; Biswas, J., Kobayashi, H., Wong, L., Abdulrazak, B., Mokhtari, M., Hutchison, D., Kanade, T., Kittler, J., Kleinberg, J.M., Mattern, F., et al., Eds.; Lecture Notes in Computer Science; Springer: Berlin/Heidelberg, Germany, 2013; Volume 7910, pp. 27–35.
8. Alsswey, A.; Al-Samarraie, H. Elderly Users' Acceptance of MHealth User Interface (UI) Design-Based Culture: The Moderator Role of Age. *J. Multimodal User Interfaces* **2020**, *14*, 49–59. [\[CrossRef\]](#)
9. Ahmed, I.; Qazi, T.F. Mobile Phone Adoption & Consumption Patterns of University Students in Pakistan. *Int. J. Bus. Soc. Sci.* **2011**, *2*, 1–10.
10. Mallerius, S.; Rossi, M.; Tuunainen, V.K. Factors Affecting the Adoption and Use of Mobile Devices and Services by Elderly People—Results from a Pilot Study. In Proceedings of the 6th Annual Global Mobility Roundtable, Los Angeles, CA, USA, 1 June 2007; pp. 1–10.
11. Gao, S.; Yang, Y.; Krogstie, J. The Adoption of Smartphones Among Older Adults in China. In *Information and Knowledge Management in Complex Systems*; Liu, K., Nakata, K., Li, W., Galarreta, D., Eds.; IFIP Advances in Information and Communication Technology; Springer International Publishing: Cham, Switzerland, 2015; Volume 449, pp. 112–122.
12. Carlsson, C.; Hyvonen, K.; Repo, P.; Walden, P. Asynchronous Adoption Patterns of Mobile Services. In Proceedings of the 38th Annual Hawaii International Conference on System Sciences, Big Island, HI, USA, 6 January 2005; IEEE: New York, NY, USA, 2005; p. 189a.
13. Hur, W.-M.; Kim, H.; Kim, W.-M. The Moderating Roles of Gender and Age in Tablet Computer Adoption. *Cyberpsychol. Behav. Soc. Netw.* **2014**, *17*, 33–39. [\[CrossRef\]](#)
14. Czaja, S.J.; Sharit, J.; Charness, N.; Fisk, A.D.; Rogers, W. The Center for Research and Education on Aging and Technology Enhancement (CREATE): A Program to Enhance Technology for Older Adults. *Gerontechnology* **2001**, *1*, 50–59. [\[CrossRef\]](#)
15. Iqbal, M.W.; Ch, N.A.; Shahzad, S.K.; Naqvi, M.R.; Khan, B.A.; Ali, Z. User context ontology for adaptive mobile-phone interfaces. *IEEE Access* **2021**, *9*, 96751–96762. [\[CrossRef\]](#)
16. Busch, P.A.; Hausvik, G.I.; Ropstad, O.K.; Pettersen, D. Smartphone Usage among Older Adults. *Comput. Hum. Behav.* **2021**, *121*, 106783. [\[CrossRef\]](#)
17. Chen, K.; Chan, A.H.S.; Tsang, S.N.H. Usage of Mobile Phones amongst Elderly People in Hong Kong. In Proceedings of the International MultiConference of Engineers and Computer Scientists, Hong Kong, China, 13–15 March 2013; Volume 2.
18. Zhou, J.; Rau, P.-L.P.; Salvendy, G. Older Adults' Use of Smart Phones: An Investigation of the Factors Influencing the Acceptance of New Functions. *Behav. Inf. Technol.* **2014**, *33*, 552–560. [\[CrossRef\]](#)
19. Roup, Z. The Use of Technology by the Elderly. *Health Sci. J.* **2010**, *4*, 10.
20. Maxwell, J.A. *Qualitative Research Design: An Interactive Approach*, 3rd ed.; Sage Publications: Thousand Oaks, CA, USA, 2013.
21. Bozan, K.; Parker, K.; Davey, B. A Closer Look at the Social Influence Construct in the UTAUT Model: An International Theory Based Approach to Investigate Health IT Adoption Patterns of the Elderly. In Proceedings of the 49th Hawaii International Conference on System Sciences (HICSS), Koloa, HI, USA, 5–8 January 2016; IEEE: New York, NY, USA, 2016; pp. 3105–3114.
22. Davis, F.D. Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Q.* **1989**, *13*, 319. [\[CrossRef\]](#)
23. Jeffrey, D. Testing the Technology Acceptance Model 3 (TAM 3) with the Inclusion of Change Fatigue and Overload, in the Context of Faculty from Seventh-Day Adventist Universities: A Revised Model. Ph.D. Thesis, Andrews University, Berrien Springs, MI, USA, 2015.
24. Gelderblom, H.; van Dyk, T.; van Biljon, J. Mobile Phone Adoption: Do Existing Models Adequately Capture the Actual Usage of Older Adults? In Proceedings of the Annual Research Conference of South African Institute of Computer Scientists and Information Technologists on-SAICSIT '10, Bela Bela, South Africa, 11–13 October 2010; ACM Press: New York, NY, USA, 2010; pp. 67–74.

25. López-Nicolás, C.; Molina-Castillo, F.J.; Bouwman, H. An Assessment of Advanced Mobile Services Acceptance: Contributions from TAM and Diffusion Theory Models. *Inf. Manag.* **2008**, *45*, 359–364. [\[CrossRef\]](#)
26. Reisdorf, B.C.; Fernandez, L.; Hampton, K.N.; Shin, I.; Dutton, W.H. Mobile Phones Will Not Eliminate Digital and Social Divides: How Variation in Internet Activities Mediates the Relationship Between Type of Internet Access and Local Social Capital in Detroit. *Soc. Sci. Comput. Rev.* **2022**, *40*, 288–308. [\[CrossRef\]](#)
27. Pai, F.-Y.; Huang, K.-I. Applying the Technology Acceptance Model to the Introduction of Healthcare Information Systems. *Technol. Forecast. Soc. Chang.* **2011**, *78*, 650–660. [\[CrossRef\]](#)
28. Lee, Y.-K.; Chang, C.-T.; Lin, Y.; Cheng, Z.-H. The Dark Side of Smartphone Usage: Psychological Traits, Compulsive Behavior and Technostress. *Comput. Hum. Behav.* **2014**, *31*, 373–383. [\[CrossRef\]](#)
29. Alsamydai, M.J. Adaptation of the Technology Acceptance Model (TAM) to the Use of Mobile Banking Services. *Int. Rev. Manag. Bus. Res.* **2014**, *3*, 13.
30. Kurniawan, S. Older People and Mobile Phones: A Multi-Method Investigation. *Int. J. Hum.-Comput. Stud.* **2008**, *66*, 889–901. [\[CrossRef\]](#)
31. Chen, S.-C.; Li, S.-H.; Li, C.-Y. Recent Related Research in Technology Acceptance Model: A Literature Review. *Aust. J. Bus. Manag. Res.* **2012**, *01*, 124–127. [\[CrossRef\]](#)
32. Alharbi, S.; Drew, S. Using the Technology Acceptance Model in Understanding Academics' Behavioural Intention to Use Learning Management Systems. *Int. J. Adv. Comput. Sci. Appl.* **2014**, *5*. [\[CrossRef\]](#)
33. Melenhorst, A.-S.; Rogers, W.A.; Bouwhuis, D.G. Older Adults' Motivated Choice for Technological Innovation: Evidence for Benefit-Driven Selectivity. *Psychol. Aging* **2006**, *21*, 190–195. [\[CrossRef\]](#) [\[PubMed\]](#)
34. Olson, K.E.; O'Brien, M.A.; Rogers, W.A.; Charness, N. Diffusion of Technology: Frequency of Use for Younger and Older Adults. *Ageing Int.* **2011**, *36*, 123–145. [\[CrossRef\]](#) [\[PubMed\]](#)
35. Chen, J.V.; Yen, D.C.; Chen, K. The Acceptance and Diffusion of the Innovative Smart Phone Use: A Case Study of a Delivery Service Company in Logistics. *Inf. Manag.* **2009**, *46*, 241–248. [\[CrossRef\]](#)
36. Putzer, G.J.; Park, Y. Are Physicians Likely to Adopt Emerging Mobile Technologies, Attitudes and Innovation Factors Affecting Smartphone Use in the Southeastern United States. *Perspect Health Inf. Manag.* **2012**, *9*, 22737094.
37. Cole, C. Looking for Information: A Survey of Research on Information Seeking, Needs, and Behavior (4th ed.). *J. Assoc. Inf. Sci. Technol.* **2017**, *68*, 2284–2286. [\[CrossRef\]](#)
38. Galesic, M.; Bosnjak, M. Effects of Questionnaire Length on Participation and Indicators of Response Quality in a Web Survey. *Public Opin. Q.* **2009**, *73*, 349–360. [\[CrossRef\]](#)
39. Bonett, D.G.; Wright, T.A. Cronbach's Alpha Reliability: Interval Estimation, Hypothesis Testing, and Sample Size Planning: Cronbach's alpha reliability. *J. Organiz. Behav.* **2015**, *36*, 3–15. [\[CrossRef\]](#)
40. Trizano-Hermosilla, I.; Alvarado, J.M. Best Alternatives to Cronbach's Alpha Reliability in Realistic Conditions: Congeneric and Asymmetrical Measurements. *Front. Psychol.* **2016**, *7*, 8. [\[CrossRef\]](#)
41. Sobon, M. E-Health Information as a Support of Health Information Technology Systems—Patients Attitudes. *Procedia Comput. Sci.* **2022**, *207*, 1723–1732. [\[CrossRef\]](#)
42. Riskinanto, A.; Kelana, B.; Hilmawan, D.R. The Moderation Effect of Age on Adopting E-Payment Technology. *Procedia Comput. Sci.* **2017**, *124*, 536–543. [\[CrossRef\]](#)